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Adaptive robot that helps you carry heavy objects like a human

Author:
Jorge Gomez Aguilar

Supervisor:
Etienne Burdet

Co-Supervisor:
Yihan Liu

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Abstract

Robots are able to perform a high variety of tasks, but their limitation in the physical interaction with humans has led to avoid this interaction. This project aims to enable a robotic arm to help a human user in the transport of a payload by integrating an impedance controller, more suited for human-robot interaction, with a human desired trajectory predictor based on haptic information. In this study, both an Unscented Kalman Filter and a Probabilistic Motion Primitives predictors are implemented in the impedance controller, testing both a fixed stiffness parameter and an adaptive stiffness control law. The developed system was experimentally validated through a human-robot collaboration reaching task with a 1 kg payload, revealing the Unscented Kalman Filter achieved a better prediction of the desired trajectory, and all the predictive impedance controllers resulted in a lower user effort.

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In addition, I must thank my parents for being there when I needed them. I could not have made it this far without you.

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Chapter 1

Introduction

1.1 Background review

Robots are increasingly being adopted by society, both in industrial spaces and outside of them [1], with direct interaction with other humans. But this increasing adoption of robots has led to reevaluate the control and human-robot interaction techniques, which used to avoid contact with people for their safety [2].

There is a need to physically interact with humans in various tasks, such as in manufacturing, rehabilitation and social robots [3, 2], but position control is too rigid for the human interaction, and force control lacks precision in the robot configuration [4]. However, impedance control may provide a better compromise, with a more human-like approach [4], by controlling both the interaction force and the position of the robot. Provided a reference trajectory and stiffness parameters (stiffness, damping and inertia), the robot interaction behaviour can be controlled to interact with the humans.

Even though this can provide the appropriate robot behaviour, this impedance control is still usually limited to a specific task, due to the need of an already known desired trajectory [2]. This control is useful in rehabilitation environments [3], where a predefined movement is repeated with the patient. However, to provide assistance in human-robot collaboration in a wide range of tasks it is necessary to determine the user's desired trajectory [2], for example in the use of an exoskeleton, or to transport an object to an unknown target.

Therefore, this project develops a predictive impedance controller, which integrates human desired trajectory prediction algorithms with an impedance controller, for a robotic arm to collaborate with the user in the transport of a payload.

The impedance control is better suited for human-robot interaction, due to its flexibility and adaptability, allowing to control both the position and the contact force simultaneously [4]. The position control [5] allows for a great tracking of a motion trajectory, useful in non-contact tasks as spray-painting [4], but it produces extreme contact forces when unexpected contact occurs. On the other hand, force control [6] allows to precisely control the contact force, however, it does not track the end-effector position. Both can be seen as extreme impedance control regimes, with a too high and too low impedance, respectively. Therefore, impedance control allows for greater flexibility in the robot behaviour, desirable for human-robot interaction [4].

However, impedance control requires a previous knowledge of the desired trajectory, which limits its application in unstructured collaboration tasks. Human desired trajectory estimation can allow the human and the robot to communicate in order to work as a team [2], extending the capabilities of impedance control. This intent prediction can be achieved from biosignals, such as electromyography [7, 8], requiring sensors in the user, or from haptic communication. The prediction of the human desired trajectory from haptic information has already been successfully used to guide a robot in a 1 DoF (Degrees of Freedom) structure, with the human tracking a

predefined trajectory [9].

This prediction of the human desired trajectory can be combined with the impedance control to allow for a flexible robot control able to consider the human intention in its control computation, obtaining information about the desired trajectory and the required motor behaviour. Therefore, the following sections delve into both the human desired trajectory estimation methods and the impedance control strategies found in the literature.

1.1.1 Desired trajectory prediction

To achieve this estimation, the methods found in the literature can be mainly divided in two categories, depending on whether they need a priori data to perform a training.

The methods without a priori data rely on a dynamics model of the system to infer the future state provided the current state. One of the most common methods is the Kalman Filter [10, 11, 2]. However, this method relies on the linearity of the system dynamics. To extend this method to address the non-linearity of the systems there are adaptations as the Extended Kalman Filter [10, 12], which approximates the non-linear functions as a first order Taylor expansion (having already been used to estimate the position of 3D objects [13], where the dynamics model is complemented by state observations from a visual system) and the Unscented Kalman Filter (UKF) [9, 14], using several points to represent the state distribution. The UKF has been able to infer the human desired trajectory and control gains using contact information [9].

From the methods requiring a priori data, or data-driven methods, the Motion Primitives [15] must be mentioned, where the prediction is done from the current state with the information from an existing dataset through the use of different algorithms such as Bayesian estimation [16], Linear Discriminant Analysis [17], Support Vector Machine [18], K-Nearest Neighbours [19] or the use of neural networks. Depending on how the Motion Primitives are established, we can distinguish between different methods. The Dynamical Motion Primitives [20] converts the experimental data into differential equations that model the movement as a system with a spring and damper summed with a nonlinear term, which contains the weights that shape the trajectory. Another example is the Probabilistic Motion Primitives (ProMP) [21, 22], in which the trajectories are the product of the basis functions and the weights. However, these weights are modeled as probability distributions, usually Normal distributions, which contain an average trajectory and its variance. This method allows to take into account the variability of human motion, which is especially relevant for human-robot interaction.

1.1.2 Robot control

Traditionally, robot control can be divided into position and force control. The core idea behind position control [5] is that the robot follows a specific trajectory (without taking into account the force needed to achieve this position), whereas in force control [4, 6] a torque is applied to the joints to create a force in the end effector (independent of its position, which cannot be guaranteed). However, the impedance controller aims to control both force and position simultaneously [23], instead of controlling these variables separately [4]. Therefore, a compliance in the robot and adaptability to the environment can be implemented, to avoid causing harm to nearby humans.

To implement this impedance control through software, without changing the mechanics of the robot [24, 23], there are three main methods: the position-based method [25], the torque-based method [26] and the model-based method [27].

In the position-based method [25], the reference trajectory is modified based on the interaction forces from the environment to obtain the desired impedance behaviour. This modified reference trajectory is fed to a traditional

position controller (see Figure 1.1). However, by depending on a position controller [5], this method is best suited for low-stiffness environments.

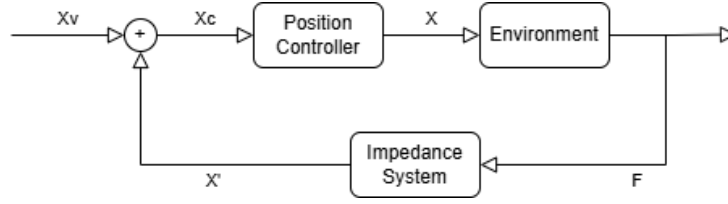


Figure 1.1: Schematic of the impedance position-based method.

In the torque-based method [26], the difference between the real trajectory and the reference trajectory is fed into the impedance equation (see equation 2.6) to obtain a torque command for the joints, which is fed into a traditional torque controller (see Figure 1.2). The dependence on the torque controller makes this method more appropriate for high-stiffness environments.

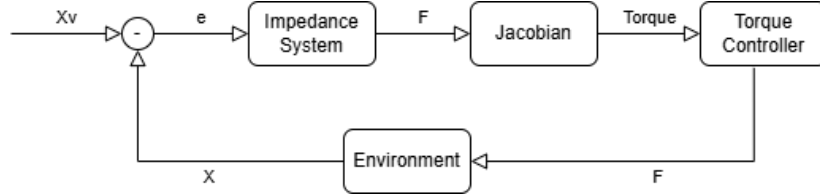


Figure 1.2: Schematic of the impedance torque-based method.

The model-based method [27] is based on the dynamical model of the robot, which relates its motion to the joints' torques. By introducing equation 2.6 into this dynamical model, the required joint torque can be obtained to follow a certain reference trajectory while maintaining the desired impedance behaviour (see Figure 1.3). However, this method can only perform the desired effect if the dynamical model is accurate enough [4].

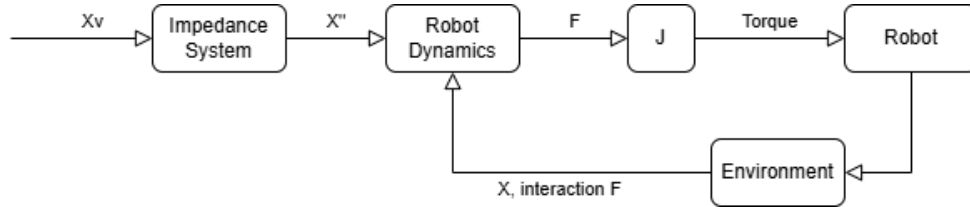


Figure 1.3: Schematic of the impedance model-based method.

In addition to these three main methods to implement the impedance control, there are some other advanced control techniques found in the literature [4], such as force-tracking impedance control, hybrid impedance control, robust impedance control and variable impedance control.

The force-tracking impedance control [28, 29, 30] aims to stabilize the contact force of the end-effector ($F(t)$) by introducing a reference contact force (F_{REF}) in the basic impedance equation, maintaining the behaviour defined by the impedance parameters, inertia (I), damping (D) and stiffness (K); as seen in equation 1.1. This modified impedance equation can be later implemented in the conventional software-based methods.

$$I * (\ddot{X} - \ddot{X}_{REF}) + D * (\dot{X} - \dot{X}_{REF}) + K * (X - X_{REF}) = F(t) - F_{REF} \quad (1.1)$$

The hybrid impedance control [31, 32] is based on the hybrid position/force control [6], which divides the taskspace in two subregions, applying a position controller or a force controller in each region to obtain a different

robot behaviour. However, this approach is limited by the accuracy of the environment modeling, as it must switch between two extreme impedance behaviours in these controllers. In the hybrid impedance control, the taskspace is divided into different subregions, and the impedance parameters change depending on the subregion in which the robot is located. Therefore, we can obtain different behaviours for different regions.

The goal of the robust impedance control [33, 34] is to maintain the desired impedance behaviour in the real world, where some external disturbance may appear. Therefore, robust impedance control is composed of different strategies, like sliding control or neural networks, to generate a signal to maintain the desired behaviour.

The variable impedance control [4, 35, 36] changes the impedance parameter online to adapt the robot behaviour to its environment. It is composed of several strategies, from adaptive control laws to machine learning techniques, including the use of deep neural networks. This adaptability is specially relevant in human-robot interaction, mimicking the ability to adapt the mechanical properties found in human neuromechanical control.

1.2 Objectives

The aim of this project is to enable a robot to collaborate with humans in versatile tasks that demand varying levels of assistance, through the development of a predictive impedance controller for the 7-DoF KINOVA Gen 3 robotic arm [37].

To achieve this goal, we can define a series of objectives:

- Design a physical interface to couple the human with the robotic arm. It should allow the human to grasp the end effector of the robotic arm comfortably, with freedom of movement in space and it should be robust enough to hold the payload in the human-robot collaboration experiments.
- Develop a predictor of human desired trajectory from the position information received by the KINOVA Gen 3 robotic arm in human-robot interaction, through the improvement of previously developed Motion Primitives and Kalman Filtering predictors. The aim is to predict the human desired trajectory online in each time step using the past observation data.
- Integrate the human desired trajectory predictor in an impedance controller for the robotic arm, to improve the performance in the human-robot interaction tasks, as the completion time and the comfort of the user, reducing the effort needed to perform the tasks.
- Improve the developed impedance controller through the use of the human effort information to adapt in real time the impedance parameters. This variable impedance control is compared to a fixed impedance controller.
- Validate the system developed through human-robot collaboration experiments, testing the accuracy of the human desired trajectory estimator and the impact of the predictive impedance controller in the tasks performance and human comfort. To test the system developed, the human and robot collaborate in the transport of a payload.

Subsequently, this report is structured into 4 chapters:

- **Chapter 1:** provides information regarding the background and objectives of the project.
- **Chapter 2:** describes the methodology employed to establish the predictive impedance system, as well as its experimental validation.

- **Chapter 3:** analyses the outcomes of the experiments performed.
- **Chapter 4:** summarizes the performance of the system developed and explores the future work.

Chapter 2

Methods

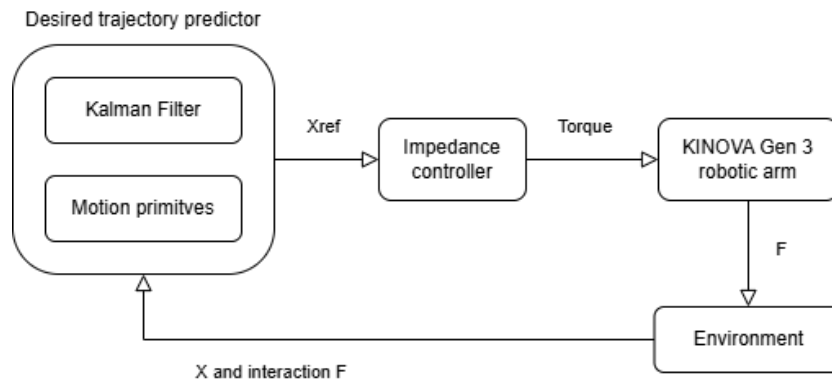


Figure 2.1: Diagram of the predictive impedance controller.

In order for the robot to be able to help a user to carry heavy objects, it should first determine where does the user want to go, as seen in Chapter 1. From the desired trajectory estimation methods discussed, both the UKF and the ProMP are used for this project. The ProMP allow to account for the variability in human movement [21], while the UKF avoids the linear dynamics limitation of the traditional Kalman Filter [14].

After this reference trajectory has been determined, an impedance controller determines the torque to be applied to the robot’s joints to achieve the intended motion with the intended impedance behaviour, as seen in Figure 2.1. In the following sections, both the estimation strategies and the impedance controller to be implemented in this project are discussed.

2.1 Desired trajectory estimation

In this project, the human hand is in direct contact to the robot’s end effector, and the KINOVA Gen 3 robotic arm has sensors to determine its own joint’s positions, velocities and torques at any timestep. This information is used to predict the human desired trajectory through the UKF and the ProMP.

2.1.1 Unscented Kalman Filter

The Kalman Filter combines the future state predicted by the model, which contains some errors due to the model’s inaccuracy, with the noisy observations from the sensors in the robot, using a Kalman gain (K) that is adapted depending on the error levels of both contributions.

The traditional Kalman Filter assumes that the true next state (z_{k+1}) is equal to the previous state (z_k) transformed by the known linear system dynamics (A) with some noise (n), as seen in equation 2.1. However, the robot is not able to perceive the true state of the system (z_k), its observation (y_k) is the true state with a distortion (C) and with some noise (w), as seen in equation 2.2.

$$z_{k+1} = A * z_k + n \quad (2.1)$$

$$y_k = C * z_k + w \quad (2.2)$$

The Kalman Filter predicts the state ($\hat{z}_{k+1}$) as the previous predicted state ($\hat{z}_k$) after the dynamics (A) are applied plus the Kalman gain (K) times the difference between the observed state (y_{k+1}) and the previous predicted state after the dynamics and distorted by the observation, as seen in equation 2.3.

$$\hat{z}_{k+1} = A * \hat{z}_k + K * (y_{k+1} - C * A * \hat{z}_k) \quad (2.3)$$

The Kalman gain represents the trust in the dynamics model against the trust in the observed state [10]. Therefore, it is adapted iteratively depending on the errors in both terms. The Kalman Filter allows to predict the next state of the system without recording any previous data, although it requires an adequate model of the system.

However, this traditional Kalman Filter is limited by the assumption of the linearity of the system dynamics [14]. To address a non-linear system transformation ($f(\hat{z}_k)$), the UKF considers the state as a probability distribution, using Gaussian Random Variables, which can be represented as a set of sample points [14]. These sample points are propagated through the non-linear transformation, and the mean and covariance of the resulting variable are used to perform the prediction in the same way as the traditional Kalman Filter (see equation 2.4). For a detailed mathematical explanation, see Section B.1.

$$\hat{z}_{k+1} = f(\hat{z}_k) + K * (y_{k+1} - \hat{y}_{k+1}) \quad (2.4)$$

2.1.2 Probabilistic Motion Primitives

Motion Primitives [15] are a way to model trajectories ($y(t)$) as a combination of basis functions ($\phi(t)$) and weights (ω), which vary for each trajectory, plus some noise (ϵ) (see equation 2.5). Therefore we can store different trajectories as a vector of their weights, which can be learnt through demonstrations.

$$y(t) = \phi(t)^T * \omega + \epsilon \quad (2.5)$$

In the ProMP the weights (ω) are probability distributions [22]. Therefore, they can represent not only the trajectory, but also the variance in the human movement, which can be used to refine the robot control in human-robot interaction [38]. In addition, they allow for movement generalization [39] and they have already been used to predict a movement trajectory from the initial movement [40, 21]. Their main drawback over other Motion Primitives methods is that they require more samples to train the weights [22]. At each timestep, the position information is used to predict the future trajectory through Bayesian estimation [16]. For a detailed mathematical explanation of this predictor, see Section B.2.

However, in order to implement a desired trajectory predictor based on the Motion Primitives it is necessary to first record some data of the desired movements, the hand position in the Cartesian space at each timestep.

2.2 Data collection

The KINOVA Gen 3 robotic arm was used to record the data for the ProMP dataset and the experimental validation, since it is equipped with the necessary joint position sensors and this recording method ensures that the resulting end effector trajectory is always inside the robot’s workspace.

However, the use of the KINOVA Gen 3 introduces some impedance in the human movements. Ideally, the user should be able to move freely while recording their hand trajectories, for the movements to be as natural as possible. To reduce this problem, a force-amplification controller has been developed to reduce the robot’s impedance in the taskspace (as seen in Appendix A).

Furthermore, a setup had to be developed to guide the user through the movements, which has been used for the predictive impedance controller experimental validation.

Finally, a human-robot interface must be designed for the user’s hand and robot end effector position to be always the same, while maintaining the comfortableness of the user to ensure the naturality of the movements. By coupling these elements, the KINOVA Gen 3 is able to record the user’s movement from the end-effector information.

2.2.1 Experimental setup

The goal is to record the human free movement trajectories between two target points in space. Therefore, a setup with clear target points is needed, without any rigid structures that might collide with the robot and interfere with the user’s movements.

A table was employed to separate the robot and the human, so that the user’s distance to the robot remains constant. An aluminum profile was placed vertically at each corner, supported by a 3D printed structure and connected through elastic cord at the top.

From these elastic cords, the targets (table tennis balls with a 40 mm diameter) hang at different heights connected through vertical elastic cords. The desired movements go from an initial position (home) to five targets (see Figure 2.2). The displacement vector in Cartesian coordinates for each movement can be seen in Table 2.1.

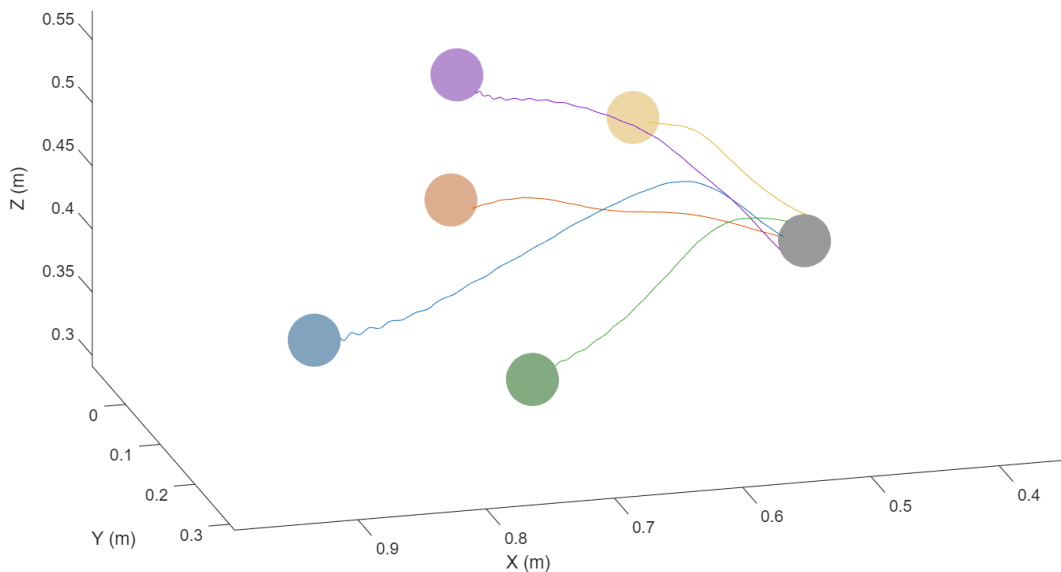


Figure 2.2: Example trajectory from the initial position to each target.

Movement	X displacement (cm)	Y displacement (cm)	Z displacement (cm)
Home to target 1	31	-22	-16
Home to target 2	20	-23	-3
Home to target 3	6	-24	0
Home to target 4	27	-9	7
Home to target 5	22	11	-8

Table 2.1: Displacement for the desired movements from the initial position

The use of lightweight targets suspended through elastic cables allows for the robot to collide with them without impedance or risk of damage to the user or the robot.

A technician is able to record all human trajectories, while the user holds the robot's end effector, through a computer connected to the KINOVA Gen 3 robotic arm, with the emergency stop always at reach in case the robot malfunctions.

2.2.2 Human-robot interface

The robot's end effector is the connection point between the human user and the robotic arm. Therefore, it must be comfortable to grip by the user and support any required payloads for the interaction experiments.

A cylindrical handle has been designed for the user to hold it in a power grip [41], to ensure the stability and power of the grip, covered by a silicon layer to avoid slipping. In addition, a tray was implemented to hold the weight in the payload experiments, with fixing points for maintaining the weight static, without introducing any unwanted dynamics, and one extension at each side to accurately reach the target positions (see Figure 2.3).

The maximum payload of the KINOVA Gen 3 robotic arm is 4 kilograms, therefore, for this project's experiments a payload of 1 kilogram is employed. In order to bring the experiments closer to the daily activities, the selected payload is a 1 kilogram package of rice.

Inside the handle are located two force sensors, in order to be able to measure the grip force of the user during the experiments.

Once the human desired trajectory predictors have been developed and the experimental setup has been decided, it is necessary to integrate the trajectory predictors with the impedance controller.

2.3 Impedance control

Impedance can be defined as the relationship between motion and force [23, 4], and therefore it can be expressed through a time differential equation by the components of inertia (I), damping (D) and stiffness (K), as seen in equation 2.6.

$$I * (\ddot{X} - \ddot{X}_{REF}) + D * (\dot{X} - \dot{X}_{REF}) + K * (X - X_{REF}) = F(t) \quad (2.6)$$

In this equation, $F(t)$ represents the contact force between the robot's end effector and the environment, the X represents the actual position of the robot and the X_{REF} represents the desired trajectory for the end-effector, which would be achieved if no interaction was present.

Since the KINOVA Gen 3 robotic arm is already developed, this project focuses on the software-based impedance control strategies. A torque-based impedance controller has already been developed for this robotic arm.

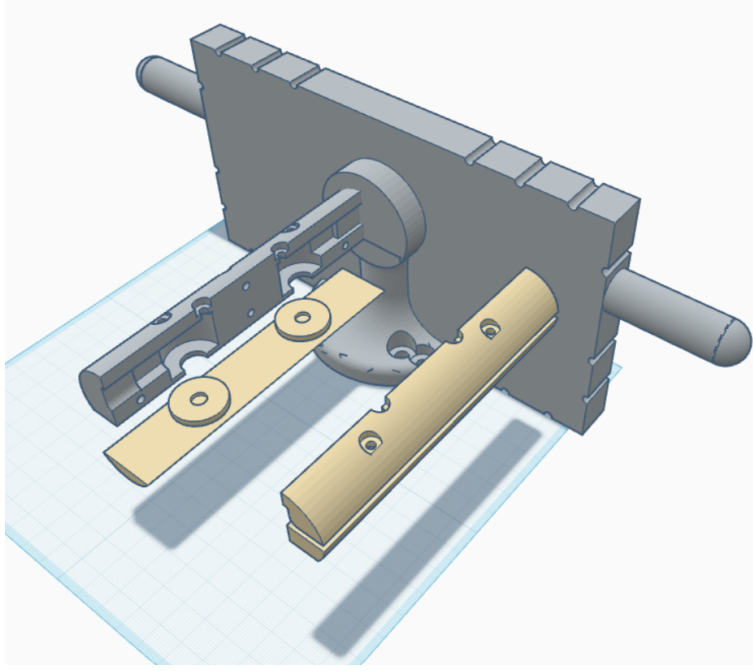


Figure 2.3: 3D model of the end effector developed for the predictive impedance experiments.

However, as previously stated in Section 1.1.2, there are several advance control strategies which can improve the current impedance controller's performance in a human-robot collaboration task. In particular, a form of variable impedance control has been implemented in the developed system.

2.3.1 Predictive impedance control

The developed predictive impedance controller takes the current state of the robot, measured through the joint position, velocity and torque sensors, and predicts the human desired trajectory in a specific time horizon using the previously discussed prediction algorithms. After this, the predicted position and velocity are used as the desired state of the impedance controller. Therefore, the robot moves towards the human desired position, with its mechanical behaviour controlled by the impedance parameters.

Even though keeping the target separated from the current position due to the prediction horizon is useful for the robot producing a torque towards the desired position, it complicates remaining motionless for the robot. Therefore, to avoid jittering caused by minimal movements from the user, a velocity threshold is implemented. If the velocity is below this threshold, the robot's position target is equal to the current position rather than the predicted position, and the velocity target is equal to zero.

Furthermore, the impedance parameters can be adapted at each timestep, aiming to reduce the user's effort. With this goal in mind, a control law has been developed to adapt the stiffness (K) according to the human effort, measured as the grasping force [42, 43] at the handle, decreasing the impedance controller's stiffness from a 1500 N/m base value proportionally to the grasping force F_{GR} (as seen in equation 2.7) until a minimum stiffness of 10 N/m is reached when the grasping force is over 49.67 N, as it was determined this was the maximum grasping force expected during the experiment. Therefore, when the user tries to take control of the robot's movements by increasing their effort, the robot becomes more compliant.

$$K = 1500 - 30 * F_{GR} \quad (2.7)$$

Finally, the predictive impedance controller developed must be experimentally validated.

2.4 Experimental validation

This project has been approved by the Imperial College Ethics Committee under the study title "Coordination Between Artificial Robotic Limbs and Human Limbs", with SETREC number 7111986. All participants gave consent to using their data for this study.

The performance of the developed system was tested through two experiments. First, a human-robot collaboration experiment to test the assistance of the predictive impedance controller in reaching tasks. In addition, the accuracy human desired trajectory predictors was tested using the data collected in the collaboration experiment.

The parameters in the controller can be seen in Table 2.2.

Parameter	Value
ProMP training dataset	300 movements
Control frequency	400 Hz
Prediction horizon	30 ms
Force-amplification stiffness matrix	$\text{diag}([0, 0, 0, 40, 40, 40])$ N/m
Impedance controller base stiffness matrix (K)	$\text{diag}([1500, 1500, 1500, 30, 30, 30])$ N/m
Impedance controller damping matrix	$2\sqrt{K}$
Predictive impedance controller's velocity threshold	0.08 m/s

Table 2.2: Controller's parameters

2.4.1 Human-robot collaboration experiment

In the human-robot collaboration experiment, the human must hold the robot's end effector handle while carrying a payload of 1 kilogram from an initial point to one out of five targets, separated in the three Cartesian dimensions, as instructed by messages displayed on a screen (see Figure 2.4). The use of a payload, within the limitations of the KINOVA Gen 3 robotic arm, aims to increase the differences in performance among controllers by increasing the effort needed to complete the task.

The experiment is divided into six trials with five different control modes. In each trial, the subject must perform 25 reaching movements, five per target goal, with a random order (see Figure 2.5). After each trial, the test subject is questioned about their perceived effort (using the Rate of Perceived Exertion [44], from 0 to 10) and their comfort level (on a scale from 0 being least comfortable to 10 being the most comfortable situation).

Five different control modes have been used:

- Force-amplification controller.
- Variable impedance ProMP predictive impedance controller: following the impedance control law seen in equation 2.7.
- Variable impedance UKF predictive impedance controller: following the impedance control law seen in equation 2.7.
- Fixed impedance ProMP predictive impedance controller: with the stiffness set in Table 2.2.
- Fixed impedance UKF predictive impedance controller: with the stiffness set in Table 2.2.



Figure 2.4: Photograph of the reaching experiment, where the user must reach the target instructed by a screen while carrying 1 kg of rice, assisted by the robot.

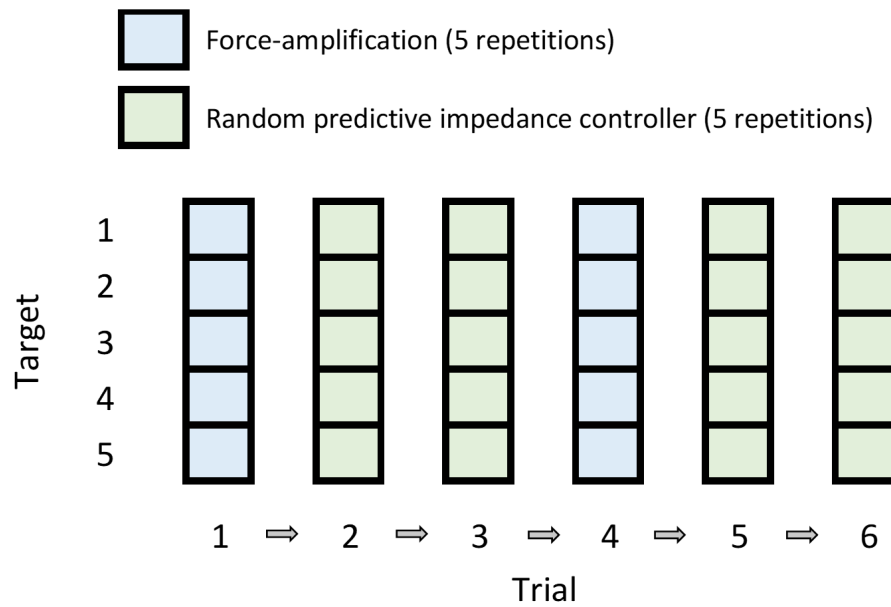


Figure 2.5: Illustration of experimental protocol, with 5 repetitions of 5 targets in a random order for 6 trials.

Before starting the first trial, there is a familiarization step, where the test subject can try the different control modes with the aim to avoid a bias in the results by the user not understanding the robot's behaviour. This step ends after five minutes or if the user verbally expresses their desire to start the first trial.

The first trial is the force-amplification controller, where the robot doesn't assist the human in the transport of the payload and therefore is used as a control for the comparison with the predictive impedance controller modes. Afterwards, two randomly selected modes are used for the next two trials. The force-amplification controller is employed for the fourth trial, to analyse whether the result differences are due to the user's experience with the robot as a whole, rather than due to the predictive impedance controller. The last two trials employ the remaining two control modes, randomly selected.

At each reaching movement, the following variables are recorded: end-effector position, user's grasping force, predicted end-effector position and interaction force. These variables are analysed afterwards to study the predictive impedance controller performance and effect on user's effort and comfort.

2.4.2 Human desired trajectory predictors

To evaluate the system developed, the performance of the human desired trajectory predictors must be assessed, both in their ability to predict a human trajectory and inside of the predictive impedance controller.

First, the predictors are evaluated independently (not affecting the next steps of the trajectory) using the force-amplification controller trajectories, recorded in the previous experiment. At each timestep, the human desired trajectory predictors predict the next second of the trajectory at 400 Hz using the position information available up to that timestep. Afterwards, this prediction is compared to the true trajectory, analysing the absolute and integrated error at 50 ms, 100 ms and 500 ms, as well as the maximum prediction horizon (the amount of time before the predicted trajectory deviates a distance threshold from the true trajectory) for a 10 cm, 5 cm and 1 cm deviation threshold.

Furthermore, the human desired trajectory predictors are evaluated in a coupled fashion, in the predictive impedance controller. At each timestep, the predictor determines a desired state in a set prediction horizon, used as the desired state of the impedance controller. Therefore, this predicted state can be compared to the true state of the end-effector at the timestep plus the prediction horizon, obtaining the error between the predicted state and the true state. However, this evaluation differs from the previous one, since the true state of the robot is affected by the robot's torque produced by the impedance controller, depending on the predicted state.

Chapter 3

Results and discussion

The human-robot collaboration experiment was performed by 15 healthy test subjects (10 male and 5 female) aged between 21 and 62 years old. For all the statistical tests, the normality of the data is assumed due to the Central Limit Theorem, and the null hypothesis is rejected if $p - value < 0.05$.

3.1 Human-robot collaboration experiment

3.1.1 Grasping force

The user's grasping force, measured at the robot's handle, can be used as a measurement of human effort [42, 43] in the reaching task.

For each movement's recording, the grasping force is averaged across each time-step. Afterwards, an N-way ANOVA test is performed with Subject, Mode and Target as independent variables. This test reveals all variables affect the grasping force, and the interaction among these variables is also significant. Therefore a post-hoc Tukey's test is performed for each variable to evaluate their effect.

For the Mode's Tukey's test (see figure 3.1), the first force-amplification trial (FA-1) yields the highest grasping force at 11.7 N, which significantly differs from the rest of the trials. The second force-amplification trial (FA-2) presents a lower grasping force (10.7 N), although significantly higher than the robot assisted control modes (ranging from 7.7 N to 8.6 N). The variable ProMP predictive controller presents a significantly higher grasping force (8.6 N) than the fixed UKF controller (7.7 N), and there are no statistically significant differences among the remaining control modes.

Regarding the Target's Tukey's test, only the third target presents a statistically significant lower grasping force than the rest of the targets.

Since the grasping force is directly proportional to the human effort for the reaching task [42, 43], it can be concluded that the predictive impedance controller reduces the effort required by the user for all the control modes, even though the variable impedance ProMP controller provides less assistance. Besides, the reduced effort in the second force-amplification trial can be explained due to the user having more experience interacting with the robot and the system as a whole.

3.1.2 Interaction force

The interaction forces at each axis recorded by the KINOVA Gen 3 robotic arm are transformed into the modulus of the interaction force vector at each timestep.

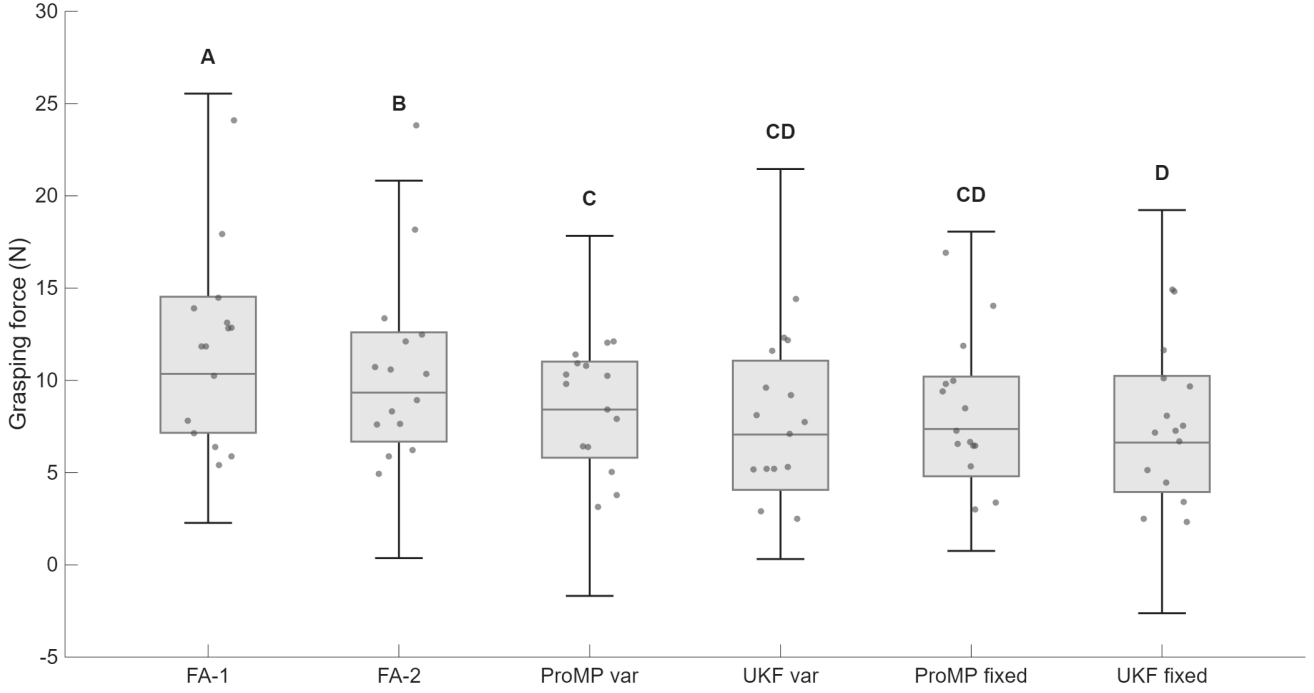


Figure 3.1: Box-plot representing the grasping force for each trial, and scatter plot for the mean grasping force for each subject in each trial. Groups not sharing a letter differ significantly (Tukey HSD, $\alpha = 0.05$), Force-amplification 1 and Force-amplification 2 trials differ from all other groups and each other at $p - value < 0.01$

For each movement's recording, the interaction force is averaged across each time-step. Afterwards, an N-way ANOVA test is performed with Subject, Mode and Target as independent variables. This test reveals all variables affect the interaction force, and the interaction among these variables is also significant. Therefore a post-hoc Tukey's test is performed for each variable to evaluate their effect.

The Mode's Tukey's test reveals there is no significant difference between both force-amplification trials (12.7 N and 12.8 N), although they are significantly lower than the robot assisted movements (ranging from 14.7 N to 15.2 N). For the remaining four control modes, there are no statistically significant differences (see Figure 3.2).

Furthermore, the N-way ANOVA followed by the post-hoc Tukey's test was performed for each axis, revealing that the interaction forces are significantly lower in the force-amplification controller (see Figure 3.3). These interaction forces means vary for axis X, from 10.1 N and 10.4 N in the force-amplification trials to values ranging from 10.9 N to 12.0 N in the predictive impedance controller trials; and for axis Y, from 3.7 N and 3.8 N in the force-amplification trials to values ranging from 4.9 N to 6.5 N in the predictive impedance controller trials.

With the reduced effort seen in Section 3.1.1, a reduced interaction force could be expected for the predictive impedance controllers. However, the lower interaction force present in the force-amplification controller results from the control algorithm present in these trials, which specifically aims to minimize the interaction force by generating a force in the robot's end effector in the same direction, as seen in Section A.3.

3.1.3 Reaching time

After taking the reaching time for each recording, an N-way ANOVA test is performed with Subject, Mode and Target as independent variables. This test reveals all variables affect the reaching time, and the interaction among these variables is also significant (except for the interaction between Subject and Target). Therefore a post-hoc

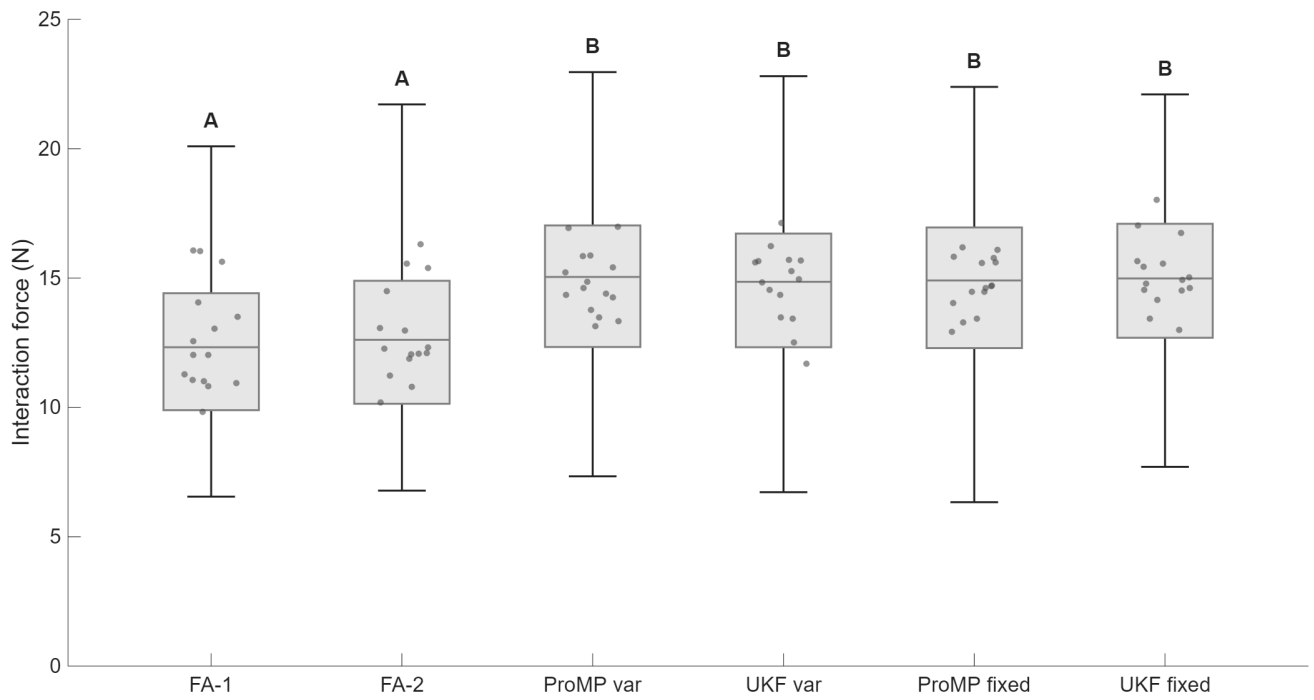


Figure 3.2: Box-plot representing the interaction force for each trial, and scatter plot for the mean interaction force for each subject in each trial. Groups not sharing a letter differ significantly (Tukey HSD, $\alpha = 0.05$).

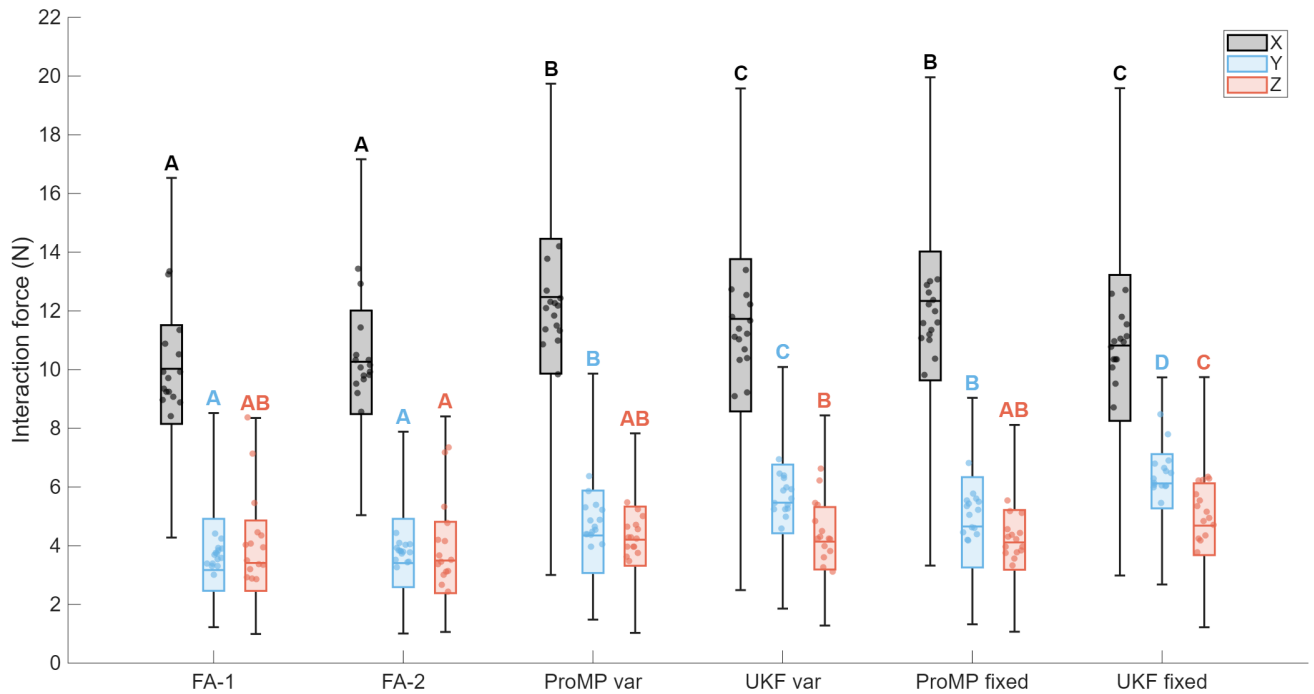


Figure 3.3: Box-plot representing the interaction force for each axis and each trial, and scatter plot for the mean interaction force for each subject in each trial and axis. For each axis, groups not sharing a letter differ significantly (Tukey HSD, $\alpha = 0.05$).

Tukey's test is performed for each variable to evaluate their effect.

In the Mode's Tukey's test, the only significant differences can be found between the second force-amplification trial (with a lower mean reaching time of 2.8 s) and the rest of the trials (ranging from 3.0 s to 3.2 s), as seen in Figure 3.4.

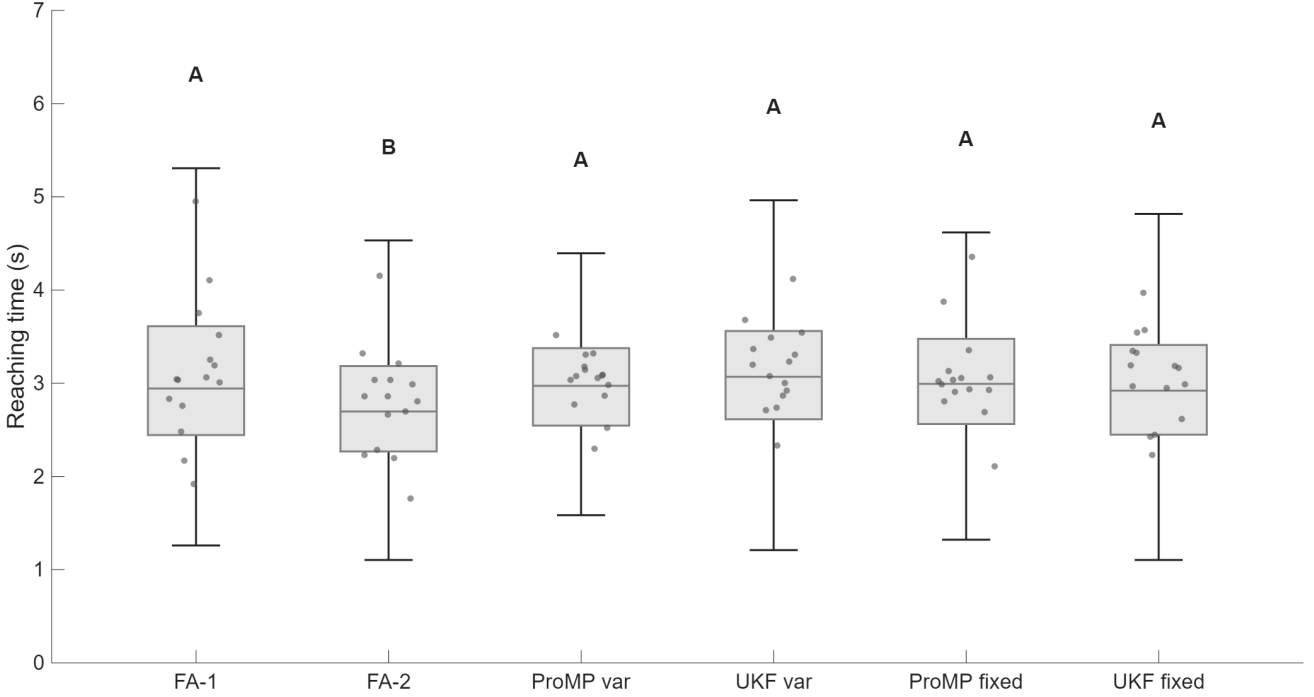


Figure 3.4: Box-plot representing the reaching time for each trial, and scatter plot for the mean reaching time for each subject in each trial. Groups not sharing a letter differ significantly (Tukey HSD, $\alpha = 0.05$).

The lower reaching time in the second force-amplification trial results from the user having more experience in the interaction with the robot (resulting in a lower human effort, as seen in Section 3.1.1). In addition, the similar reaching time in all robot assisted movements could be due to the prediction horizon. Increasing this prediction horizon would increase the distance between the predicted target and the current position, increasing the robot's torque for the same stiffness value, and therefore increasing the system's velocity to reach the target. However, a high movement speed could be uncomfortable for less experienced users, producing a trade-off between reaching time and user comfort.

3.1.4 Perceived effort and comfort

For the recorded perceived effort and comfort asked after each trial, a MANOVA test was performed with the Mode as the independent variable, and both the perceived effort and comfort as dependent variables. Although the mean perceived effort is lower (from 5.7 and 5.2 in the force-amplification controller to values ranging from 4.2 to 4.6 in the predictive impedance controllers) and the comfort is higher for the robot assisted trials compared to the force-amplification trials (from 5.4 and 5.7 in the force-amplification controller to values ranging from 6.2 to 6.5 in the predictive impedance controllers), these differences are not statistically significant.

The lack of statistically significant differences for these variables could be because of the reduced number of test subjects and the low weight of the payload, due to the robot's limitations (as seen in Section 2.2.2).

3.2 Human desired trajectory predictors

The human desired trajectory predictors must be evaluated, both at determining the desired trajectory from a human independently performing the reaching task, and while the user is coupled with the robot assistive system.

Regarding the independent prediction, for each movement's recording, all predictor variables (horizons, absolute errors and integrated errors) are averaged across each time-step. Then a MANOVA test was performed, with the Predictor as the independent variable. As it yielded significant differences between the predictors, a post-hoc two-sample t-test with paired samples was performed for each variable, with a Bonferroni correction $\alpha = 0.05/9$.

This test shows that all variables have statistically significant differences. The UKF predictor presents a higher time horizon for all thresholds than the ProMP predictor (from 0.81 s to 0.55 s for the 10 cm threshold, from 0.75 s to 0.54 s for the 5 cm threshold, and from 0.49 s to 0.41 s for the 1 cm threshold) as seen in Figure 3.5 A.

Besides, the UKF predictor presents both a lower absolute error and integrated error than the ProMP predictor for 500 ms (from 1.8 cm to 3.6 cm in the absolute error, and 1.6 m to 2.4 m in the integrated error), 100 ms (from 1.7 mm to 1.9 mm in the absolute error, and 3.8 cm to 4.4 cm in the integrated error) and 50 ms (from 0.82 mm to 0.92 mm in the absolute error, and 1.1 cm to 1.3 cm in the integrated error); as seen Figures 3.5 B and C.

On the other hand, for the coupled prediction, the prediction error is averaged across each time-step for each recording (discarding the force-amplification trials). After this, an N-way ANOVA test is performed with Subject, Mode and Target as independent variables. This test reveals that only the Mode significantly affects the prediction error.

Therefore, a post-hoc Tukey's test is performed for the Mode, which only presents significant differences in the variable stiffness ProMP controller, with a lower prediction error (2.4 mm) than both the UKF controllers (4.1 mm for the variable UKF and 4.4 mm for the fixed UKF), as seen in Figure 3.5 D.

With regards to the predictor evaluation, the UKF yielded better results for the independent prediction, although the ProMP predictor could be improved by increasing the training dataset. On the other hand, the prediction while coupled in the predictive impedance controller presents a lower prediction error in the variable ProMP controller, which could be due to the effect of the prediction in the movement of the end effector.

Therefore, it can be concluded that the UKF produces better results in the current architecture, although the ProMP could have more potential to improve by increasing the training dataset.

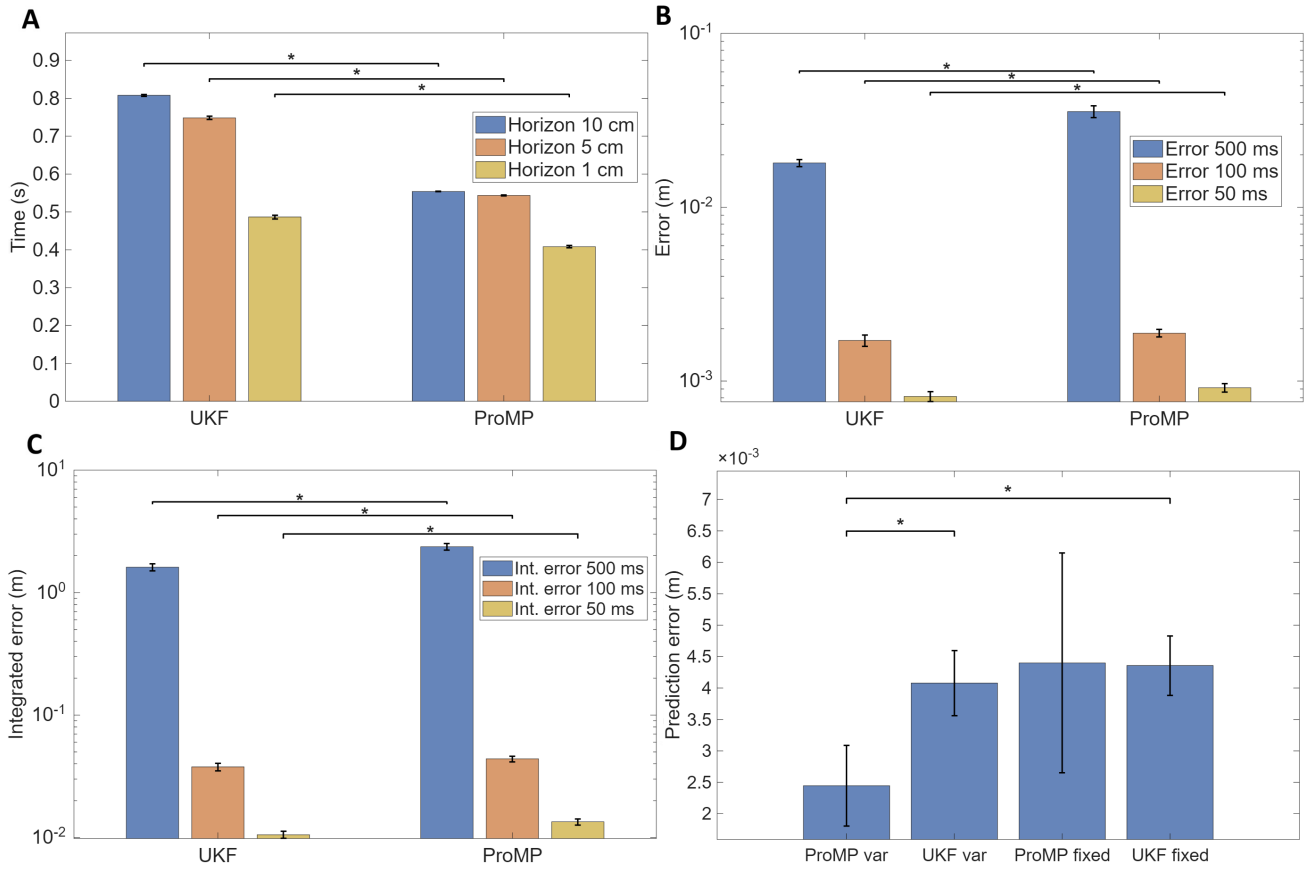


Figure 3.5: **A:** Mean horizon for each predictor with a 95% confidence interval, for a threshold of 10 cm, 5 cm and 1 cm. **B:** Mean absolute error for each predictor with a 95% confidence interval, for 500 ms, 100 ms and 50 ms. **C:** Mean integrated error for each predictor with a 95% confidence interval, for 500 ms, 100 ms and 50 ms. **D:** Error for each predictor in the predictive impedance controller with a 95% confidence interval.

Chapter 4

Conclusions

The present project developed a predictive impedance controller, assisting a human user in a reaching task with a payload in three spatial dimensions by determining their desired trajectory through haptic information. The prediction by the UKF and ProMP was compared, as well as the use of a traditional impedance controller and an adaptive impedance controller.

Firstly, the human desired trajectory predictors have been developed, guiding an existing impedance controller in the reaching task. For the human-robot collaboration, a custom end effector has been created to facilitate the coupled interaction and to carry the payload. Through the reaching experiment it could be shown that:

- The UKF predictor presents a better prediction than the ProMP predictor, although it did not significantly impact the predictive impedance controller's performance.
- The predictive impedance controllers result in a lower user effort, but there are no significant differences among the controllers (except for the variable impedance ProMP controller), and the interaction force is not as low as in the force-amplification controller.
- There is no perceived difference by the user in comfort or effort among the controllers.

Overall, it can be concluded that the objectives proposed for this project have been achieved and the system developed has succeeded in reducing the user's effort in the reaching task with a payload. However, there are several limitations to this study, such as:

- The payload weight limitation imposed by the KINOVA Gen 3 robotic arm, since an increased payload could increase the differences among controllers.
- The limited number of participants in the experimental validation, as well as the gender inequality.
- The simpleness of the adaptive impedance control law, resulting in no significant difference with the fixed impedance.
- The fixed parameters shown in Table 2.2, which could be optimized in further studies.

Thus, the future work may focus on:

- Testing the effect of the parameters shown in Table 2.2 on the robot's performance. In particular, increasing the ProMP training dataset could provide a lower prediction error.

- Improving the adaptive impedance controller's law to adapt the controller to each user's preferences in real time.
- Performing the experimental validation with more subjects, to increase the statistical power of the obtained results.
- Testing the developed architecture in a different hardware. The benefits range from being able to increase the payload in a different robotic arm, to finding different applications for the predictive impedance controller in human-robot collaboration tasks, such as in the use of exoskeletons.

Appendix A

Data collection algorithm

For the human movements to be as natural as possible while coupled with the robotic arm, it is necessary to control the mechanical behaviour of the robot's end effector, for it to have as low inertia, damping and stiffness as possible. This results in a minimization of the interaction force (f_{ext}).

Admittance control creates an appropriate mechanical behaviour [45] by measuring the external force in the end-effector and generating a motion command [46, 47]. Due to the relation between force and motion, it is considered dual, and even sometimes a subtype, of the Impedance control [46, 48, 49]. The difference between these control modes resides in the causality of the controller [48, 49], since the admittance control takes as input the external force and produces a motor command, while the impedance control takes the motion error as input and produces a resulting force.

As in the Impedance controller [4], the goal is to generate a mechanical behaviour by relating both the external force at the end effector (f_{ext}) and motion displacement from a desired state (p_d) through a desired inertia (I_d), damping (D_d) and stiffness (K_d) in the taskspace [48, 47], as seen in equation A.1:

$$I_d * (\ddot{p} - \ddot{p}_d) + D_d * (\dot{p} - \dot{p}_d) + K_d * (p - p_d) = f_{ext} \quad (\text{A.1})$$

By measuring the external force, it is possible to obtain the required motion to produce the desired behaviour by solving for the acceleration ($\ddot{p}$). Afterwards, this motion command can be implemented in the robot through a torque or position controller.

Therefore, the end effector acceleration ($\ddot{p}$) can be computed provided the external force from equation A.1 as A.2:

$$\ddot{p} = I_d^{-1} * (f_{ext} - D_d * (\dot{p} - \dot{p}_d) - K_d * (p - p_d)) + \ddot{p}_d \quad (\text{A.2})$$

A.1 Torque-based admittance controller

The real behaviour of the robot is not described by the desired impedance behaviour equation A.1, but by the robot's dynamic equation A.3 in the jointspace [48, 47].

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau_m + \tau_{ext} \quad (\text{A.3})$$

The total torque in the joints, the combined torque produced by the motors (τ_m) and the external torque (τ_{ext}), must be equal to the combined torque produced by the mass matrix ($M(q)$) and the joints acceleration ($\ddot{q}$), the

Coriolis and centripetal effect ($C(q, \dot{q})\dot{q}$) and the gravitation effect ($g(q)$). This equation can be transformed to the taskspace through [A.4](#):

$$M_x \ddot{p} + C_x \dot{p} + g_x = f_m + f_{ext} \quad (\text{A.4a})$$

$$M_x = J^{\dagger T} M(q) J^{\dagger} \quad (\text{A.4b})$$

$$C_x = J^{\dagger T} (C(q, \dot{q}) - M(q) J^{\dagger} \dot{J}) J^{\dagger} \quad (\text{A.4c})$$

$$g_x = J^{\dagger T} g(q) \quad (\text{A.4d})$$

$$f_m = J^{\dagger T} \tau_m \quad (\text{A.4e})$$

$$f_{ext} = J^{\dagger T} \tau_{ext} \quad (\text{A.4f})$$

The pseudoinverse Jacobian ($J^{\dagger}$) is used due to the Jacobian matrix not being a square matrix. From equation [A.4](#), we can express the end effector force (f_m) in the taskspace as [A.5](#).

$$f_m = M_x \ddot{p} + C_x \dot{p} + g_x - f_{ext} \quad (\text{A.5})$$

Therefore, the acceleration ($\ddot{p}$) computed in [A.2](#) can be introduced in [A.5](#) to produce the desired mechanical behaviour, as seen in [A.6](#).

$$f_m = M_x (I_d^{-1} (f_{ext} - D_d(\dot{p} - \dot{p}_d) - K_d(p - p_d)) + \ddot{p}_d) + C_x \dot{p} + g_x - f_{ext} \quad (\text{A.6})$$

Finally, the end effector taskspace force (f_m) can be transformed into the jointspace through [A.7](#). The KI-NOVA Gen 3 robotic arm is able to follow this torque command (τ_m) through low-level torque control, producing the desired end-effector effect.

$$\tau_m = J^T f_m \quad (\text{A.7})$$

Therefore, the robot's behaviour can be controlled through the taskspace desired mechanical constants, stiffness (K_d), damping (D_d) and inertia (I_d) [[47](#)]; as well as the desired motion ($p_d, \dot{p}_d, \ddot{p}_d$).

Since the goal is for the user's movement to be as free as possible and to remain motionless if there is no external force, both the desired acceleration ($\ddot{p}_d$) and velocity ($\dot{p}_d$) must be equal to zero. Besides, to reduce the impedance of the robot, the damping and inertia must be as low as possible [[46](#)].

Due to the fact that the taskspace coordinates are a combination of the Cartesian directions and the rotations, the stiffness must be equal to zero in the Cartesian directions (since the robot must not return to any location) and high in the rotation components, to maintain the end effector orientation constant. Finally, the desired position is irrelevant in the Cartesian directions, as the stiffness is zero, but must maintain the specific orientation for the end effector in the rotation components.

The torque-based admittance controller was implemented in the KINOVA Gen 3 robotic arm with a 400 Hz control frequency and a butterworth filter [50] with a 12 Hz cutoff frequency for the measured external force signal. However, when testing its performance, severe oscillations were observed in the robot end effector. The origin of this problem is the lowering of the desired inertia (I_d), creating a high gain for the control output (equation A.6) which results in instability at the end effector, a common challenge in admittance control [46]. Even though increasing the desired inertia reduces the oscillation, it increases the impedance for the user, worsening the performance of the system.

A.2 Position-based admittance controller

The position-based admittance controller produces a jointspace position command as its output for a low-level position controller to move the robot's joints. Therefore, the taskspace acceleration obtained in equation A.2 must be converted into a jointspace position command.

The first step is to obtain the taskspace velocity command by integrating the taskspace acceleration command ($\ddot{p}_u$) through the Euler method [51], for an integration step dt , as seen in equation A.8.

$$\dot{p}_u = \dot{p} + \ddot{p}_u * dt \quad (\text{A.8})$$

Afterwards, the taskspace velocity command can be converted to the jointspace through the pseudoinverse Jacobian (equation A.9).

$$\dot{q}_u = J^\dagger \dot{p}_u \quad (\text{A.9})$$

Finally, the jointspace position command can be obtained by integrating again through the Euler method, using the jointspace velocity, as seen in A.10. This jointspace position can be followed by the KINOVA Gen 3 robotic arm with a low-level position controller.

$$q_u = q + \dot{q}_u * dt \quad (\text{A.10})$$

However, the use of equation A.3 also results in high acceleration when reducing the desired impedance for the free movement, affecting the control output. Thus, instability and oscillations appear in the end-effector, as seen in the torque-based admittance controller.

A.3 Force-amplification controller

The goal of this controller is to compensate the impedance of the robot, so the human can move as freely as possible. This problem can be reformulated into increasing the user's force, so that the robot's resistance is negligible. Therefore, another approach to the controller design is force amplification, found sometimes in the literature as human extenders [52]. This control algorithm is often employed in exoskeletons to increase the user's strength [53, 54].

The key to force-amplification control is the robot producing a force (f_m) proportional to the user's force (f_{ext}) by some gain (α) [52], as seen in equation A.11. Therefore, if the human tries to move upwards and encounters a resistance, a force upward is measured, and the robot applies a higher force (provided $\alpha > 1$) against the resistance, reducing the effort the user needs to apply.

$$f_m = \alpha f_{ext} \quad (\text{A.11})$$

In addition, other terms can be introduced to the control law for compensating the Coriolis and centripetal effect as well as the gravitation effect (see equation [A.12](#)).

$$f_m = \alpha f_{ext} + C_x \dot{p} + g_x \quad (\text{A.12})$$

Finally, to ensure the orientation is maintained, a stiffness component can be introduced, with values different from zero only in the rotation components. The final control law obtained can be seen in equation [A.13](#), where α is set to 1.5 to obtain an almost free movement control.

$$f_m = \alpha f_{ext} + C_x \dot{p} + g_x - K_d(p - p_d) \quad (\text{A.13})$$

Appendix B

Mathematical explanation of the human desired trajectory predictor

B.1 Unscented Kalman Filter

In this project, the state (z) is defined as the position (p), velocity (v), an acceleration factor (L) and the endpoint position or goal (g); as seen in equation B.1.

$$z = \begin{bmatrix} p \\ v \\ L \\ g \end{bmatrix} \quad (\text{B.1})$$

The predicted state ($\hat{z}_{k+1}$) is updated according to the non-linear dynamic model ($f(\hat{z}_k)$) and the observations from the environment (y_{k+1}) compared to an estimate of this observation ($\hat{y}_{k+1}$), as seen in equation B.2.

$$\hat{z}_{k+1} = f(\hat{z}_k) + K * (y_{k+1} - \hat{y}_{k+1}) \quad (\text{B.2})$$

The non-linear dynamic model updates at each time-step (dt) the position (p) based on the velocity (v), and the velocity based on the acceleration factor (L) and the difference between the goal and the current position, while keeping constant the acceleration factor and the goal (as seen in equations B.3).

$$p_{k+1} = p_k + v_k * dt \quad (\text{B.3a})$$

$$v_{k+1} = v_k + L_k * (g_k - p_k) * dt \quad (\text{B.3b})$$

$$L_{k+1} = L_k \quad (\text{B.3c})$$

$$g_{k+1} = g_k \quad (\text{B.3d})$$

In the UKF the state is modelled as a probability distribution, which is represented by several points (separated by a factor depending on the state covariance P). These points are transformed by the dynamic model and

reconstructed into the predicted state, with its mean ($f(\hat{z}_k)$) and covariance (P), using some weights (ω_m and ω_c) and the process noise (Q), as seen in equation B.4.

$$f(\hat{z}_k) = \sum_{i=1}^N \omega_m^i * f(\hat{z}_k^i) \quad (\text{B.4a})$$

$$P = Q + \sum_{i=1}^N \omega_c^i * (f(\hat{z}_k^i) - f(\hat{z}_k)) * (f(\hat{z}_k^i) - f(\hat{z}_k))^T \quad (\text{B.4b})$$

Once an estimate of the future state has been obtained through the non-linear model, the difference between this estimate and the observed state is taken into account (as seen in equation B.2). The observed state is comprised of the current position and velocity in Cartesian coordinates, and an estimation of the goal through the solution of a minimum-jerk trajectory equation, by fitting the current trajectory to a polynomial to estimate the current acceleration, and then assuming the hand will decelerate from the current position to a stop (as seen in equation B.5). The acceleration factor is not measured from the robot, therefore, the observation matrix (C) does not take into account this acceleration factor (see equation B.6). The points from the state, transformed by the non-linear dynamics, are used to create an estimate of the observation ($\hat{y}_{k+1}$), as seen in equation B.7.

$$g = p + k_1 * v * T + k_2 * a * T^2 \quad (\text{B.5})$$

$$C = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \quad (\text{B.6})$$

$$\hat{y}_{k+1} = \sum_{i=1}^N \omega_m^i * C * f(\hat{z}_k^i) \quad (\text{B.7})$$

Finally, the estimated state from the non-linear dynamics is combined with the observation through the Kalman gain (K), computed by taking into account the observation noise (R), as seen in equations B.8.

$$P_{zz} = R + \sum_{i=1}^N \omega_c^i * (C * f(\hat{z}_k^i) - \hat{y}_{k+1}) * (C * f(\hat{z}_k^i) - \hat{y}_{k+1})^T \quad (\text{B.8a})$$

$$P_{xz} = \sum_{i=1}^N \omega_c^i * (f(\hat{z}_k^i) - f(\hat{z}_k)) * (C * f(\hat{z}_k^i) - \hat{y}_{k+1})^T \quad (\text{B.8b})$$

$$K = P_{xz} * P_{zz}^{-1} \quad (\text{B.8c})$$

After this, the Kalman gain is used to update the state's covariance, for the sampling of the next iteration's state (see equation B.9).

$$P = P - K * P_{zz} * K^T \quad (\text{B.9})$$

However, to obtain a state prediction further in time an additional step is needed. The state can be propagated

through the non-linear dynamics (see equation B.10) as many times as needed to obtain a forecast at a specific horizon.

$$\hat{z}_{k+1} = f(\hat{z}_k) \quad (\text{B.10})$$

B.2 Probabilistic Motion Primitives

In the human robot interaction, we can record the user's trajectories while performing a reaching task to different goals. The end effector position in the jointspace is recorded through the controller discussed in Appendix A and the joint velocities are computed using the Savitzky-Golay filter. Both the joint position and velocity are encoded through the motion primitives, and therefore the equation 2.5 can be rewritten as B.11:

$$\begin{bmatrix} y(t) \\ \dot{y}(t) \end{bmatrix} = \begin{bmatrix} \phi(t)^T & 0 \\ 0 & \dot{\phi}(t)^T \end{bmatrix} * \begin{bmatrix} \omega_{pos} \\ \omega_{vel} \end{bmatrix} + \epsilon = H * \omega + \epsilon \quad (\text{B.11})$$

Both sets of weights can be learnt through the ridge regression, using a regularization coefficient $\lambda > 0$. For each demonstration i , the weights can be computed as seen in equation B.12.

$$\omega_{pos}^i = (\phi^T \phi + \lambda I)^{-1} \phi^T y^i \quad (\text{B.12a})$$

$$\omega_{vel}^i = (\dot{\phi}^T \dot{\phi} + \lambda I)^{-1} \dot{\phi}^T \dot{y}^i \quad (\text{B.12b})$$

Assuming the weights of the motion primitives follow a Gaussian distribution, its mean and variance can be easily computed from the weights obtained for each demonstration (ω^i), as seen in equation B.13.

$$\mu_\omega = \frac{1}{N} \sum_{i=1}^N \omega^i \quad (\text{B.13a})$$

$$\Sigma_\omega = \frac{1}{N-1} \sum_{i=1}^N (\omega^i - \mu_\omega)(\omega^i - \mu_\omega)^T \quad (\text{B.13b})$$

Thus, we can predict the human desired trajectory from the weight distribution, using the partial movement observed. This prediction can be done through Bayesian estimation [16], using the previously defined variables as well as an arbitrary established standard deviation of the observations (σ_y).

The goal is to compute a posterior weight distribution (with a mean and variance) from the prior (μ_ω and Σ_ω), updated by the observation of the trajectory (see equation B.14).

$$\Sigma_{post} = (\Sigma_\omega^{-1} + \frac{1}{\sigma_y^2} H^T H)^{-1} \quad (\text{B.14a})$$

$$\mu_{post} = \Sigma_{post} (\Sigma_\omega^{-1} \mu_\omega + \frac{1}{\sigma_y^2} H^T y) \quad (\text{B.14b})$$

Finally, we have obtained a trajectory prediction encoded with the motion primitives that becomes the reference trajectory for the robot controller, using the posterior weights as seen in equation B.15.

$$\hat{y}(t) = \phi(t)^T * \mu_{post} \tag{B.15}$$

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